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Research paper

Edge-based multimodal sensor data fusion with Vision-Language-Action (VLA) model for real-time autonomous vehicle accident avoidance

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ABSTRACT

Autonomous driving (AD) systems relying solely on onboard sensors may fail to detect distant or occluded hazards, potentially causing preventable collisions; however, existing transformer-based Vehicle-to-Everything (V2X) approaches either lack effective multimodal fusion and reasoning or struggle to meet real-time performance requirements under complex traffic conditions. This paper proposes the **R**eal-time **E**dge-based **A**utonomous **C**o-pilot **T**rajectory planner (REACT), a V2X-integrated trajectory optimization framework for AD based on a fine-tuned lightweight Vision–Language–Action (VLA) model explicitly grounded in low-level control generation. REACT integrates infrastructure-provided hazard alerts with onboard sensor data, capturing surrounding traffic dynamics and vehicle intents through visual embeddings, interpreting numerical data from symbolic inputs, and employing contextual reasoning to predict control-aware residual action sequences. To enable robust real-time deployment on edge devices, REACT employs a Residual Trajectory Fusion (RTF) module that predicts discrete residual controls and fuses them with a nominal plan, together with edge-adaptation strategies to reduce model complexity and improve inference efficiency. Evaluated on the DeepAccident benchmark, REACT achieves state-of-the-art performance, a 84.9% collision rate reduction, a 50.6% Video Panoptic Quality, and a 0.57-second inference latency on the Jetson AGX Orin. Ablation studies validate the contribution of each input modality, architectural component, and edge adaptation strategy. These results highlight the effectiveness of lightweight control-grounded VLA reasoning for real-time cooperative planning on edge platforms and underscore the potential of language-guided residual control generation for improving traffic safety.

1. Introduction

Road traffic crashes remain a critical global issue, claiming approximately 1.19 million lives and injuring up to 50 million individuals annually. They are the leading cause of death among individuals aged 5 to 29 (World Health Organization, 2023). Among the contributing factors, impaired, distracted, or careless driving stands out as the primary cause, accounting for 29.2% of all accidents and resulting in over 3000 fatalities each year in the U.S. alone (National Highway Traffic Safety Administration, 2023; Centers for Disease Control and Prevention, 2023). This significant toll underscores the urgent need for autonomous driving (AD) and Advanced Driver-Assistance Systems (ADAS) to help mitigate human error and reduce distraction-related collisions (Zhang et al., 2025a). Although these technologies hold promise for enhancing road safety, their capabilities are often limited by the perception range of onboard sensors, which can delay or prevent the detection of occluded or distant hazards. As a result, ADAS and AD systems may fail to respond effectively to unexpected threats, highlighting the necessity for a more proactive and context-aware driving framework (Huang et al., 2023; Li et al., 2025).

To overcome the safety limitations of isolated onboard sensing, connected autonomous vehicle (CAV) networks leverage Vehicle-to-Everything (V2X) communication to integrate diverse transportation agents, including vehicles, roadside units (RSUs), and traffic management entities. This collaborative framework enhances system-wide situational awareness, enabling individual vehicles to perceive beyond their immediate sensor range, including blind spots and broader traffic conditions, thereby supporting more proactive and informed decision-making (Abdi et al., 2024). Despite its potential, large-scale V2X deployment faces notable challenges in prevailing perception methodologies. Models such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Vision Transformers (ViTs) (Khan et al.,

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2022) often suffer from limited generalization, requiring extensive labeled datasets and struggling with novel or unseen scenarios (Raghu et al., 2021; Cui et al., 2025). Moreover, non-attention-based methods excel primarily in pattern recognition but fail to meet the high-level contextual reasoning required for complex real-world driving (Wang et al., 2023a; Fu et al., 2024; Dou et al., 2023). Besides the aforementioned limits, integrating multimodal information remains a substantial challenge due to the architectural complexity and limitations of existing model designs (Joshi et al., 2021; Truhn et al., 2024; Wu et al., 2023).

Recent advances in Vision–Language Models (VLMs) have demonstrated that extending Large Language Models (LLMs) with visual perception enables powerful multimodal reasoning beyond conventional sensor-driven pipelines (Brown et al., 2020; Wang et al., 2025; Bilal et al., 2025). By jointly modeling visual context and semantic knowledge, VLMs support high-level understanding of complex, dynamic environments, including intent inference, rule compliance, and rare-event reasoning, which are critical for autonomous driving under occlusion, dense traffic, and long-tail scenarios (Wang et al., 2024a; Yang et al., 2025). Moreover, their inherent ability to fuse heterogeneous visual and symbolic inputs facilitates scalable multimodal integration and improves generalization through zero-shot and few-shot learning, reducing reliance on exhaustive labeled datasets (Li et al., 2024; Saha et al., 2024; Lei et al., 2024). However, most existing VLM-based autonomous driving approaches remain perception- or decision-oriented, producing high-level semantic interpretations or future state predictions without directly grounding model outputs in executable vehicle actions (Ahn et al., 2022; Driess et al., 2023).

Recent frontier research has shifted toward Vision–Language–Action (VLA) models that tightly couple multimodal reasoning with executable driving actions. CoVLA (Arai et al., 2025) establishes a large-scale vision–language–action dataset for autonomous driving, enabling direct supervision of action grounding. Building on this paradigm, OpenDriveVLA (Zhou et al., 2025b) and AutoVLA (Zhou et al., 2025a) demonstrate end-to-end driving systems that map visual and linguistic inputs to low-level control through large VLA models with adaptive reasoning and reinforcement fine-tuning. By unifying perception, language reasoning, and action prediction, VLA models form a principled foundation for real-time, safety-critical planning in complex V2X-enabled traffic environments.

Building upon these insights, this research introduces a Real-time Edge-based Autonomous Co-pilot Trajectory planner (REACT), a V2X-enabled framework leveraging a lightweight VLA model to improve transportation safety. REACT comprises five core modules: (1) sensing data acquisition, (2) data structuring, (3) task alignment, (4) task projection enhancement, and (5) Residual Trajectory Fusion (RTF). The sensing module gathers multimodal inputs, including onboard camera frames, navigation goals, ego vehicle states, and hazard alerts from RSUs. Visual data is converted into Bird's-Eye View (BEV) maps that capture road geometry and nearby road users. To support action-grounded reasoning, REACT stacks BEV maps from the current moment and a prior timestamp, enabling the VLA model to infer scene dynamics and driving intent relevant to control generation. The data structuring module filters noise, extracts necessary information, and validates the RSU hazards. The task alignment stage synchronizes multimodal inputs and encodes both visual and symbolic information into optimized prompts. The task projection enhancement module then passes those prompts into a lightweight VLA model, which directly predicts the changes of a temporally dense sequence of low-level vehicle control commands, including steering angles and longitudinal accelerations. Finally, the RTF module reconstructs the predicted control sequence, enabling proactive collision avoidance.

A notable feature of REACT is its edge-based deployment, which demonstrates real-time responsiveness and minimizes delays caused by network latency and bandwidth limitations with edge computing (Chen et al., 2023; Baller et al., 2021). Large VLA models typically incur high computational overhead due to joint perception, reasoning, and action generation, often exceeding the capabilities of edge devices (Chen et al., 2023). To address this challenge, REACT introduces innovative optimization strategies that combine prompt engineering (Marvin et al., 2023) with parameter-efficient fine-tuning techniques (Chen et al., 2024a; Luo et al., 2024), achieving a model reaction latency reduction of 93.67% without sacrificing planning performance. In terms of model architecture, REACT incorporates the RTF module that mitigates the high latency and low accuracy of direct VLA control prediction by estimating residual control adjustments over a nominal plan. RTF clips and fuses these residuals to generate optimized sequential maneuvers, and experiments confirm its necessity for stable real-time performance. Experimental results indicate that the combination of REACT's optimization strategies and the RTF module improves system efficiency by 95.25%, enhances control prediction accuracy and trajectory fidelity, and demonstrates greater training stability.

In summary, REACT addresses key limitations of conventional perception methodologies in V2X systems by deploying a lightweight VLA model on edge devices. The primary contributions of this research are as follows:

- **Architectural Contribution:** This paper introduces REACT, a V2X-integrated autonomous driving framework built upon a customized VLA model that directly generates low-level vehicle control sequences for safer and more context-aware autonomous driving.
- **Methodological Innovation:** This study proposes a lightweight VLA-based multimodal reasoning pipeline that jointly grounds perception and action, enabling the prediction of steering and acceleration command sequences. Furthermore, an innovative RTF module is introduced to improve both system computation efficiency and accuracy for collision risk mitigation.
- **Edge Optimization:** REACT incorporates targeted edge adaptation strategies, including prompt optimization and parameter-efficient fine-tuning, that streamline model complexity and enable real-time control inference on resource-constrained edge devices without compromising planning quality.
- **Practical Applicability:** The proposed system is validated under diverse traffic scenarios, environmental conditions, and infrastructure configurations, demonstrating robust decision-making and control performance in realistic V2X-enabled deployment settings.

2. Literature review

Large vision–language and VLA models are increasingly recognized for their potential to unify perception, semantic reasoning, and action generation in autonomous driving. Recent studies on multimodal sensorimotor control, contextual end-to-end driving, lightweight real-time perception, and edge deployment of VLMs further show that practical autonomous systems require not only multimodal understanding but also efficient, deployable, and control-oriented intelligence (Munir et al., 2023; Azam et al., 2024; Zhang et al., 2025b; Rashid et al., 2025). To contextualize this work, this review examines three complementary dimensions: VLA-based modeling for driving behavior, multimodal V2X integration for cooperative reasoning, and edge-oriented adaptation of large multimodal models to support stable, resource-efficient control generation in real-world driving systems.

2.1. VLA models for driving behavior

VLA models extend vision–language learning by grounding multimodal perception and language reasoning directly into executable driving actions, enabling a unified framework that bridges scene understanding, intent reasoning, and vehicle control for autonomous driving (Jiang et al., 2025b; Hu et al., 2026). Some efforts in this direction emphasize datasets and supervision, recognizing that effective action grounding requires large-scale paired vision, language, and action annotations. CoVLA (Arai et al., 2025) establishes one of the first comprehensive VLA datasets for autonomous driving, providing synchronized video, natural-language descriptions, and driving actions across diverse traffic scenarios, thereby enabling supervised learning and standardized evaluation of VLA models. Building upon such data foundations, recent research advances reasoning and reinforcement fine-tuning to improve the coherence and safety of VLA outputs. AutoDrive-R^2 (Yuan et al., 2025) explicitly incentivizes chain-of-thought (CoT) reasoning and self-reflection during training, while reinforcement learning objectives are employed to align language-guided reasoning with safe and consistent driving behaviors.

In parallel, a growing body of work explores generative and planning-oriented VLA models, framing decision-making as a conditional generative process to better handle multimodality and uncertainty. DiffVLA (Jiang et al., 2025a) integrates vision–language guidance with diffusion-based planning, allowing the model to sample diverse yet plausible maneuver candidates under ambiguous or interactive driving conditions. Complementing these approaches, end-to-end VLA architectures aim to directly map multimodal inputs to driving actions through large-scale models. OpenDriveVLA (Zhou et al., 2025b) proposes hierarchical alignment between perception, language, and action to enable end-to-end driving, while AutoVLA (Zhou et al., 2025a) further incorporates adaptive reasoning and reinforcement fine-tuning to enhance robustness and generalization across scenarios.

Recognizing the computational burden of large VLA models, recent studies focus on computation and efficiency to enable real-time deployment. FastDriveVLA (Cao et al., 2025) introduces reconstruction-aware token pruning to reduce inference costs while preserving driving-critical information, demonstrating that an efficiency-aware design is essential for practical VLA-based autonomous driving systems.

Despite rapid advances, existing VLA methods largely focus on trajectory- or plan-level outputs and often lack the ability to reason over multi-source, multi-modal V2X information to generate stable, executable control commands at sufficiently high temporal resolution. Moreover, direct long-horizon control prediction remains inefficient and unstable, as residual control formulations and controlled fusion mechanisms for stabilizing sequential maneuvers are rarely explored.

2.2. Vision-language models for V2X

The integration of VLMs with V2X communication is an emerging research direction aimed at enhancing cooperative perception and decision-making (Yu et al., 2025; Jalil et al., 2025). While explicit VLA formulations for V2X remain limited, existing studies on VLM- and LLM-based V2X systems lay important groundwork by demonstrating how semantic reasoning over multi-source, multi-modal V2X data can improve situational awareness and planning. Early efforts such as V2X-VLM (You et al., 2026) introduce an end-to-end cooperative driving framework that fuses onboard and infrastructure sensing into a unified semantic representation using VLMs, improving trajectory planning robustness under complex and occluded conditions. By incorporating contrastive learning for multimodal alignment, these approaches highlight the value of semantic fusion for cooperative perception, though their outputs remain largely trajectory- or decision-level rather than action-grounded.

Subsequent studies further emphasize high-level semantic reasoning and abstraction in V2X-enabled driving. Frameworks such as the Advanced Driving Agent with Multimodal LLMs (Chen and Lu, 2024) and AutoReward (Han et al., 2024) leverage multimodal inputs and CoT prompting to support distributed decision-making and adaptive policy learning through language-guided reasoning. SenseRAG (Luo et al., 2025) extends this line of work by introducing retrieval-augmented generation within a V2X context, constructing a dynamic environmental knowledge base that interprets real-time sensor data and V2X messages via language-based reasoning. These systems demonstrate scalable and flexible semantic fusion across heterogeneous V2X sources but primarily focus on reasoning and planning rather than generating executable control commands.

At a broader level, LLM-centric V2X frameworks further illustrate the potential of language-based coordination among connected agents. AgentsCoMerge (Hu et al., 2024) proposes an LLMs-driven multi-agent collaboration system for ramp merging through semantic communication, while V2X-LLM (Wu et al., 2025) integrates LLMs into connected vehicle corridors to reason over large-scale V2X data streams such as BSMs and SPaT messages. Although these approaches do not explicitly incorporate visual grounding or action generation, their architectures are naturally extensible toward VLA paradigms by incorporating vision-conditioned inputs and action-oriented outputs.

Overall, existing VLM- and LLM-based V2X studies demonstrate the feasibility and benefits of semantic reasoning over multi-source V2X data; yet they largely stop short of closing the loop from cooperative perception and reasoning to stable, high-frequency, and executable vehicle control. Recent V2X cooperative driving benchmarks and challenges similarly emphasize cooperative temporal perception and end-to-end planning, but still leave open the problem of converting multimodal V2X reasoning into directly executable low-level control (Hao et al., 2025). This gap motivates the development of VLA-based V2X frameworks, such as REACT, that explicitly reason over heterogeneous V2X modalities and translate cooperative semantic understanding into conductible control commands suitable for real-world autonomous driving.

2.3. Edge-based large models

Edge computing enables computational intelligence to be deployed close to data sources such as vehicles and RSUs, supporting responsive, privacy-preserving, and bandwidth-efficient processing in AD systems (Chen and Ran, 2019; Chen et al., 2023). As large vision–language and VLA models continue to scale, adapting them for real-time inference on resource-constrained edge platforms has become a prerequisite for practical deployment, particularly for safety-critical driving tasks that require timely action generation rather than offline analysis. Recent work on resource-constrained VLM deployment further confirms that hardware-aware scheduling, quantization, and lightweight visual encoding are essential for deploying multimodal reasoning models on edge platforms (Rashid et al., 2025).

Researchers have explored architectural strategies to support edge-level execution of large multimodal models, offering insights relevant to VLA deployment. EdgeVLA (Budzianowski et al., 2025) investigates simplified visuomotor architectures by removing autoregressive decoding and reducing model size, demonstrating that lightweight designs can still support time-sensitive action inference. Hybrid paradigms such as EdgeCloudAI (Ghasemi et al., 2024) distribute computation between edge and cloud, enabling lightweight edge-side perception and preprocessing while offloading deeper semantic reasoning, a strategy that can be extended to VLA systems that balance local action generation with global context reasoning. VEANET (Zhong et al., 2025) further explores dynamic edge assistance through migrating AI agent twins across RSUs, illustrating how edge infrastructure can continuously support autonomous vehicles in motion.

Complementary efforts focus on model compression and systematic evaluation under resource constraints. MobileAIBench (Murthy et al., 2024) provides standardized benchmarks to analyze the effects of quantization and runtime optimization on large language and multimodal models, highlighting trade-offs between accuracy and efficiency. Lightweight VLMs such as MiniCPM-V (Yao et al., 2024), MobileVLM, and MobileVLM V2 (Chu et al., 2023, 2024) demonstrate that competitive multimodal reasoning can be achieved with reduced parameter counts, while methods like Align-KD (Feng et al., 2024) preserve cross-modal alignment through knowledge distillation. Although

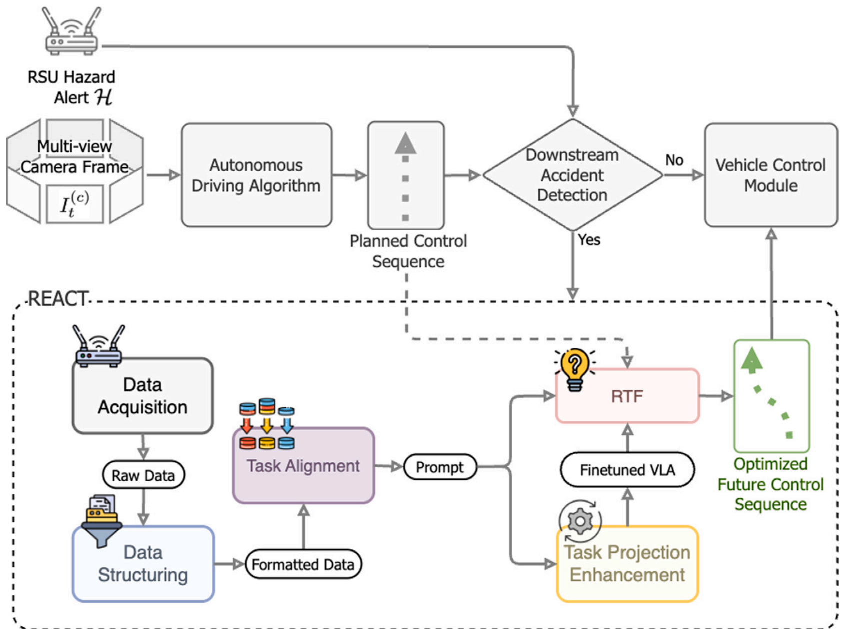

Fig. 1. The overview of the REACT.

these approaches primarily target perception and reasoning tasks, they establish key techniques for scaling VLA models to edge platforms.

Overall, existing edge-focused studies demonstrate promising directions for deploying large multimodal models under resource constraints, yet most remain centered on perception, dialogue, or general robotic assistance rather than AD control. In particular, limited attention has been given to enabling edge-deployable VLA systems that reason over multi-source, multi-modal V2X inputs and generate stable, high-frequency, executable control commands. This gap motivates the design of REACT, which adopts lightweight VLA principles to support cooperative reasoning and reliable control generation in edge-based V2X-enabled AD systems.

3. Methodology

The REACT framework enables real-time, proactive driving assistance by jointly reasoning over multi-source, multi-modal V2X information and onboard sensing to generate stable, executable vehicle control commands through a lightweight VLA model. This section presents the overall system architecture, control-centric design choices, and inference workflow of REACT, as illustrated in the system overview.

3.1. Framework overview

The proposed framework is designed to operate alongside an existing AD system that produces a nominal future control plan based on onboard perception and planning modules (Chen et al., 2024b). REACT is activated only when a potential hazard is detected by a downstream RSU. As illustrated in Fig. 1, REACT runs in parallel to the AD stack and refines the nominal driving behavior by reasoning over multi-source, multi-modal V2X information to generate stable and executable control commands through five interconnected modules.

The **Data Acquisition** module collects raw multimodal inputs, including sensor data from onboard sensors and vehicle systems, as well as hazard alerts received via V2X communication from nearby RSUs. These inputs form the foundational data stream for subsequent processing.

The **Data Structuring** module selects useful information from raw data, filters noise, processes visual data into BEV feature maps, and converts them into structured, model-compatible representations. This improves the quality and consistency of the data passed to downstream modules.

In the **Task Alignment** module, structured visual features and symbolic information are jointly encoded into optimized prompts. These prompts compactly describe the traffic context, cooperative hazard cues, vehicle states, and control objectives and are specifically designed to support efficient and accurate inference within a VLA model.

The **Task Projection Enhancement** module uses CoT to supervise fine-tuning (SFT) of a lightweight VLA with paired textual and visual inputs. CoT-based SFT improves the model's ability to interpret prompts by learning intermediate reasoning steps, leading to clearer task understanding and stable predictions during inference.

Finally, the **Residual Trajectory Fusion (RTF)** module enables the VLA to predict Gaussian Mixture Model (GMM) (Rasmussen, 1999) encoded residual control adjustments instead of full control sequences, which are decoded, clipped, and fused with the nominal plan to produce stable, executable control commands without trajectory reconstruction.

Together, these modules enable REACT to reason over cooperative V2X information and onboard perception while generating high-frequency, stable, and conductible control commands suitable for real-time deployment. Detailed descriptions of each module and their implementation are provided in the following sections, and the overall architecture of REACT is illustrated in Fig. 2.

3.2. System configuration

To establish a clear foundation for evaluating the proposed REACT framework in typical V2X safety scenarios, this paper considers a driving situation in which an autonomous ego vehicle encounters a downstream collision that is occluded by preceding traffic and therefore cannot be directly perceived by onboard sensors. Instead, the ego vehicle receives cooperative hazard information via V2X communication, which must be interpreted and acted upon through control-level intervention. Table 1 summarizes the definitions and explanations for each symbol used in this paper.

All vehicle positions are defined in an ego-centric coordinate frame $F_E = \{\hat{\mathbf{x}}_{\text{forward}}, \hat{\mathbf{y}}_{\text{left}}, \hat{\mathbf{z}}_{\text{up}}\}$, with the ego vehicle located at the origin $(0, 0, 0)$. Velocities and yaw angles are expressed in an RSU-aligned reference frame to remain consistent with infrastructure-provided measurements.

Table 1
Summary of mathematical symbols.

Symbol	Definition	Explanation
F_E	Ego-centric frame	Coordinate frame centered at the ego vehicle $(0,0,0)$.
F_R	RSU reference frame	Infrastructure-aligned coordinate frame used for velocity and yaw.
$\mathbf{R}_{ER}$	Frame rotation matrix	Rotation matrix from RSU frame F_R to ego frame F_E.
$\hat{\mathbf{x}}_{\text{forward}}$	Ego x-axis	Unit vector pointing forward in the ego frame.
$\hat{\mathbf{y}}_{\text{left}}$	Ego y-axis	Unit vector pointing leftward in the ego frame.
$\hat{\mathbf{z}}_{\text{up}}$	Ego z-axis	Unit vector pointing upward in the ego frame.
k	Timestep index	Discrete time index in the planning horizon.
$\mathcal{K}$	Time index set	Set of all discrete timesteps.
$\mathbf{s}_k$	Ego kinematic state	Ego velocity and yaw: $\mathbf{s}_k = [v_{x,k}, v_{y,k}, \psi_k]^\top$.
$v_{x,k}, v_{y,k}$	Ego velocities	Longitudinal and lateral velocities in RSU frame.
ψ_k	Ego yaw angle	Heading angle of ego vehicle.
Δt	Time interval	Sampling interval between timesteps.
$\mathcal{H}$	RSU hazard alert	Collision location and time $\{x_h, y_h, z_h, t_h\}$.
x_h, y_h, z_h	Hazard position	Hazard location relative to RSU.
t_h	Hazard time	Time to or since collision occurrence.
$\mathcal{N}$	Navigation set	Ordered list of route waypoints.
$\mathbf{g}_j$	Waypoint	Spatial navigation reference point j.
S_{ego}	Ego state history	Short-horizon ego kinematic history.
$I_t^{(c)}$	Camera image	Image from camera c at time t.
$\mathbf{B}_t$	BEV feature map	Bird's-Eye View representation at time t.
$\mathbf{B}_t^{\text{aug}}$	Augmented BEV	BEV with bounding-box overlay highlighting road users.
$\mathbf{g}_h^R$	Hazard position (RSU frame)	Hazard location $(x_h, y_h, z_h)^\top$ expressed in RSU frame F_R.
$\mathbf{g}_{\text{ego}}^R$	Ego position (RSU frame)	Ego vehicle position expressed in RSU frame F_R.
$\mathbf{g}_h^E$	Hazard position (ego frame)	Hazard location transformed into ego-centric frame F_E.
θ_h	Relative hazard direction	Bearing of the hazard relative to ego heading in ego frame.
Δt_h	Time-to-hazard	Remaining time until predicted collision, $\Delta t_h = t_h - t_{\text{now}}$.
$\mathcal{M}_m$	Object mask	Binary BEV mask for object $m \in \mathcal{O}$.
$\mathcal{U}$	Control sequence	Sequence of steering and acceleration pairs $\{(\delta_k, a_k)\}$.
δ_k	Steering angle	Steering command at timestep k.
a_k	Acceleration	Longitudinal acceleration at timestep k.
$\Delta\mathbf{u}_n$	Control residual	Residual control $(\Delta\delta_n, \Delta a_n)$.
K	GMM components	Number of Gaussian components in control clustering.
π_k	GMM weight	Mixture weight of component k.
$\boldsymbol{\mu}_k$	GMM mean	Prototype residual control of component k.
$\boldsymbol{\Sigma}_k$	GMM covariance	Covariance of residual control distribution.
$\hat{\mathbf{c}}$	Predicted classes	Sequence of residual control class indices.
$\hat{c}_n$	Control class	Predicted residual control class at step n.
$\mathbf{u}_k^{\text{nom}}$	Nominal control	Original planned control at timestep k.
$\hat{\mathbf{u}}_k$	Optimized control	Control after RTF fusion and clipping.
$\mathcal{P}$	VLA prompt	Full multimodal prompt for VLA inference.
$\mathcal{P}_{\text{Inst}}$	Instruction template	Fixed control-oriented instruction prompt.
P_t	Textual prompt	Encoded hazard, horizon, and control context.
P_v	Visual prompt	BEV-based visual input to the VLA.
$\mathcal{R}$	CoT trace	Chain-of-Thought used only during SFT.
N	Control horizon	Number of future control steps predicted.

At discrete timestep $k \in \mathcal{K}$, the ego vehicle state is defined as Eq. (1), where $(v_{x,k}, v_{y,k})$ denote the longitudinal and lateral velocities and ψ_k denotes the yaw angle.

$$\mathbf{s}_k = [v_{x,k},\ v_{y,k},\ \psi_k]^\top, \tag{1}$$

The RSU broadcasts hazard information $\mathcal{H}$ as a concise alert in Eq. (2) (Zheng et al., 2024; He et al., 2023), indicating the position (x_h, y_h, z_h) of the collision relative to the RSU and its predicted occurrence timestamp $t_h \geq 0$, where $t_h = 0$ denotes an accident that has already occurred, and $t_h > 0$ indicates the accident is anticipated to occur in t_h seconds.

$$\mathcal{H} = \{x_h,\ y_h,\ z_h,\ t_h\}, \tag{2}$$

The on-board route planner supplies a finite ordered list of spatial waypoints without fixed timestamps that the ego vehicle must sequentially reach. The navigation set $\mathcal{N}$ is denoted by Eq. (3), where $\mathcal{J} = \{1, \ldots, M\}$ indexes the M waypoints and each vector $\mathbf{g}_j = (x_j^{\text{ref}},\ y_j^{\text{ref}},\ z_j^{\text{ref}})^\top$ specifies the desired longitudinal, lateral, and vertical coordinates of waypoint j in the RSU-centered frame F_R. The final element $\mathbf{g}_M$ corresponds to the route destination or the next high-level navigation milestone.

$$\mathcal{N} = \left\{\mathbf{g}_j\right\}_{j\in\mathcal{J}}^M = \left\{\left(x_j^{\text{ref}},\ y_j^{\text{ref}},\ z_j^{\text{ref}}\right)\right\}_{j=1}^M, \tag{3}$$

3.3. Data acquisition

The *Data Acquisition* module aggregates all multimodal inputs required for downstream VLA reasoning, serving as the interface between raw sensing, cooperative V2X information, and the REACT inference pipeline. At each discrete timestep $k \in \mathcal{K}$, the collected input set $\mathcal{D}_k$ is denoted as Eq. (4), where $\mathcal{I}_k$ represents visual observations.

$$\mathcal{D}_k = \left\{\mathcal{I}_k,\ \mathbf{s}_k,\ \mathcal{H},\ \mathcal{N}\right\}, \tag{4}$$

The ego vehicle captures synchronized multi-view camera images $\mathcal{I}_k$, denoted as Eq. (5), where c indexes camera views. These images are later transformed into BEV representations for spatial reasoning. Ego-vehicle kinematic information is obtained from the Controller Area Network (CAN) bus and represented by the state vector $\mathbf{s}_k$.

$$\mathcal{I}_k = \{I_k^{(c)} \mid c \in \mathcal{C}\}, \tag{5}$$

Through V2X communication, the ego vehicle receives RSU broadcast $\mathcal{H}$, describing the location and anticipated time of a downstream collision. In addition, the onboard route planner provides $\mathcal{N}$ defining the intended route. Together, $\mathcal{D}_k$ constitutes a unified multi-source, multi-modal input space that enables REACT to reason over local perception and the cooperative V2X context.

3.4. Data structuring

The *Data Structuring* module transforms heterogeneous raw inputs into structured representations suitable for VLA inference. Given the acquired input set $\mathcal{D}_k$ in Eq. (4), this module performs three parallel processing pipelines: visual feature structuring, hazard information validation, and ego vehicle intention and status extraction.

Visual Processing

Multi-view camera images $I_t^{(c)}$ are transformed into ego-centric BEV feature maps to support spatial and temporal reasoning. To encode short-horizon motion, two BEV representations are constructed at time t and $t - \Delta t$ using 3D reconstruction and spatial pooling (Mao et al., 2023), shown as Eq. (6). The paired BEV features capture static layout and implicit motion cues without explicit velocity estimation.

$$\mathbf{B}_t^{\text{now}} = \text{MaxPool}\Big(\mathcal{F}_{\text{bev}}(I_t^{(c)})\Big), \quad \mathbf{B}_{t-\Delta t}^{\text{past}} = \text{MaxPool}\Big(\mathcal{F}_{\text{bev}}(I_{t-\Delta t}^{(c)})\Big). \tag{6}$$

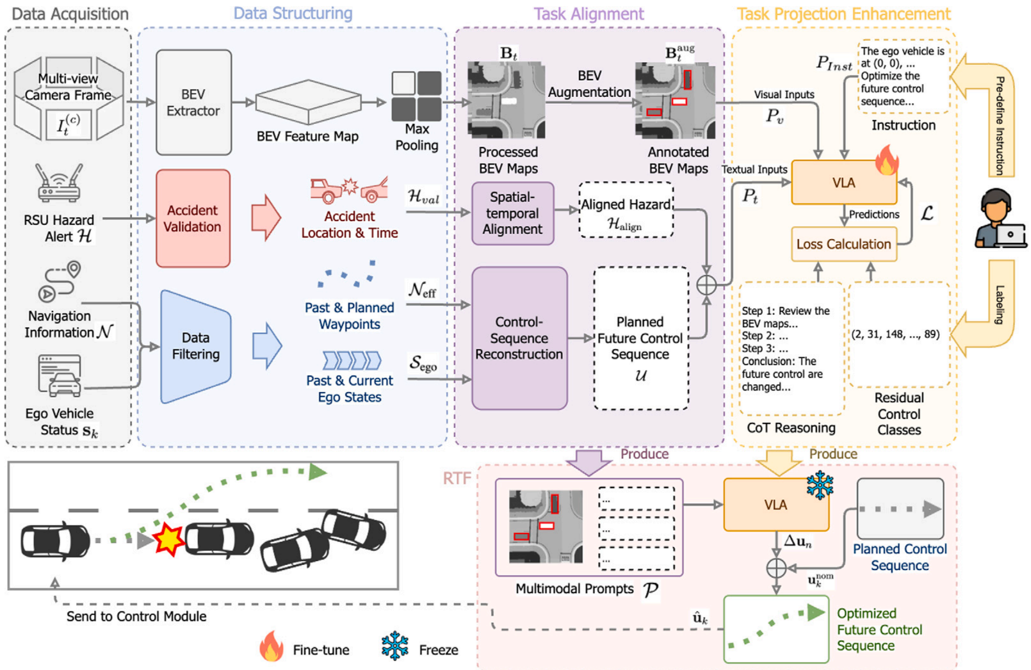

Fig. 2. The architecture of the REACT.

Accident Information Validation

To ensure the reliability $\mathcal{H}$, a rule-based validation procedure is applied prior to downstream reasoning. The validity of a hazard alert is determined by Eq. (7), which enforces two physically grounded consistency conditions: the hazard must be spatially downstream of the ego vehicle's planned route and temporally relevant to the current decision window. The validated hazard $\mathcal{H}_{\mathrm{val}}$ is then defined as Eq. (8). Invalid or outdated alerts are discarded to prevent erroneous control interventions. This validation step is intended as a conservative plausibility gate for structured RSU alerts rather than a standalone hazard classifier; the final control decision remains jointly grounded in the validated hazard cue, paired BEV observations, and the nominal control sequence during multimodal reasoning.

$$\mathrm{Valid}(\mathcal{H}) = \mathbb{1}\left[\mathbf{g}_M \cdot \hat{\mathbf{x}}_{\mathrm{up}} > x_h \;\wedge\; |t_h - t_{\mathrm{now}} < \delta_t\right], \tag{7}$$

$$\mathcal{H}_{\mathrm{val}} = \mathrm{Valid}(\mathcal{H}) \cdot \mathcal{H}. \tag{8}$$

Ego Intention & Status Extraction

To provide compact and control-relevant symbolic inputs for downstream VLA reasoning, the navigation intent and ego vehicle status are temporally filtered to retain only information relevant to short-horizon decision-making. Given the $\mathcal{N}$, only a limited future segment $\mathcal{N}_{\mathrm{eff}}$ is preserved, shown as Eq. (9), where M' denotes the number of retained future waypoints, effectively encoding the near-term route intent.

$$\mathcal{N}_{\mathrm{eff}} = \left\{\mathbf{g}_j \in \mathcal{N} \;\middle|\; j \in \{k, k+1, \dots, k+M'\},\; M' \ll M\right\}, \tag{9}$$

Similarly, ego vehicle kinematic states are temporally truncated to capture short-term motion context while avoiding redundant historical information. The ego state history is defined as Eq. (10), where K denotes the history length in seconds.

$$\mathcal{S}_{\mathrm{ego}} = \left\{\mathbf{s}_{k'} \;\middle|\; k' \in \{k-K, \dots, k\}\right\}, \tag{10}$$

These compact and curated representations serve as the input to the *Task Alignment* module, ensuring that all symbolic inputs are contextually relevant and lightweight for efficient reasoning.

3.5. Task alignment

The *Task Alignment* module transforms structured visual and symbolic inputs into representations aligned for VLA inference. It consists of two sequential components: BEV feature augmentation and control-sequence reconstruction, followed by prompt generation.

BEV Augmentation

Given the max-pooled BEV feature map $\mathbf{B}_t = \{\mathbf{B}_t^{\mathrm{now}}, \mathbf{B}_{t-\Delta t}^{\mathrm{past}}\}$, road users are highlighted by overlaying bounding boxes corresponding to detected object regions. Let $\mathcal{M}_m \in \{0,1\}^{H\times W}$ denote the binary mask of object $m \in \mathcal{O}$, and let $\partial\mathcal{M}_m$ denote the boundary of its minimal enclosing bounding box. The augmented BEV map is defined as Eq. (11), where $\mathbf{1}_{\partial\mathcal{M}_m}$ is an indicator function that equals one on the bounding box boundary and zero elsewhere, and λ is a fixed highlighting intensity. This overlay operation visually emphasizes road users while preserving the original BEV features and global road structure.

$$\mathbf{B}_t^{\mathrm{aug}} = \mathbf{B}_t + \sum_{m\in\mathcal{O}} \lambda \cdot \mathbf{1}_{\partial\mathcal{M}_m}, \tag{11}$$

Hazard Alignment

The RSU broadcasts $\mathcal{H}_{\mathrm{val}}$ in the RSU reference frame F_R. To align this information with ego-centric perception, the hazard location is transformed into the ego frame F_E via a rigid transformation in Eq. (12), where $\mathbf{g}_h^R = (x_h, y_h, z_h)^\top$, $\mathbf{g}_{\mathrm{ego}}^R$ is the ego position in F_R, and $\mathbf{R}_{ER}$ denotes the rotation from F_R to F_E. The relative hazard direction with respect to the ego heading is then computed as Eq. (13). The temporal proximity of the hazard is obtained as $\Delta t_h = t_h - t_{\mathrm{now}}$, indicating the remaining time until the predicted collision. The aligned hazard representation $\mathcal{H}_{\mathrm{align}} = (\theta_h, \Delta t_h)$ is subsequently used in prompt construction for control reasoning.

$$\mathbf{g}_h^E = \mathbf{R}_{ER}\big(\mathbf{g}_h^R - \mathbf{g}_{\mathrm{ego}}^R\big), \tag{12}$$

$$\theta_h = \arctan 2\big(y_h^E,\, x_h^E\big). \tag{13}$$

Control-Sequence Reconstruction

Most public autonomous driving datasets provide waypoint-based trajectories without explicit low-level control annotations. When such annotations are unavailable, REACT reconstructs steering and acceleration sequences from waypoint positions. Given waypoints $\{(x_k, y_k)\}_{k=1}^N$

from $\mathcal{S}_{\text{ego}}$, accelerations are computed via finite differences in Eq. (14). The longitudinal acceleration is computed by projecting $\mathbf{a}_k$ onto the velocity direction, shown as Eq. (15).

$$\mathbf{v}_k = \frac{(x_{k+1} - x_{k-1},\, y_{k+1} - y_{k-1})}{t_{k+1} - t_{k-1}}, \quad \mathbf{a}_k = \frac{\mathbf{v}_{k+1} - \mathbf{v}_{k-1}}{t_{k+1} - t_{k-1}}. \tag{14}$$

$$a_k = \frac{\mathbf{a}_k^\top \mathbf{v}_k}{\|\mathbf{v}_k\|}. \tag{15}$$

The vehicle heading ψ_k is obtained from $\mathcal{S}_{\text{ego}}$, and the yaw rate $\dot{\psi}_k$ is computed by temporal differentiation. The curvature $\kappa_k = \dot{\psi}_k / \|\mathbf{v}_k\|$ yields the steering command δ_k as Eq. (16), where L denotes the vehicle wheelbase.

$$\delta_k = \arctan(L\,\kappa_k), \tag{16}$$

The reconstructed control sequence is summarized as $\mathcal{U}$, which is used for training and alignment when ground-truth control commands are unavailable.

$$\mathcal{U} = \{(\delta_k, a_k)\}_{k=1}^{N}, \tag{17}$$

3.6. Task projection enhancement

Prompt Design

REACT designs a control-oriented prompt $\mathcal{P}$ to guide VLA reasoning toward stable residual control prediction. Each prompt consists of a fixed task instruction $\mathcal{P}_{\text{Inst}}$, multimodal inputs, and a CoT trace $\mathcal{R}$ for SFT. The textual prompt P_t encodes the validated hazard, prediction horizon, control bounds, and nominal control sequence, while the visual prompt P_v provides the augmented BEV observations, shown in Eq. (18), where $\mathcal{R}$ is included only for CoT-SFT and omitted at inference.

$$\mathcal{P} = \left[\mathcal{P}_{\text{Inst}} \,\|\, P_t(\mathcal{U}, \mathcal{H}_{\text{align}}) \,\|\, P_v(\mathbf{B}_t^{\text{aug}}, \mathbf{B}_{t-\Delta t}^{\text{aug}}) \,\|\, \mathcal{R}\right], \tag{18}$$

The prompt is optimized for VLA understanding. An example used in our finetuning process is:

```
BEV_t-1:<image> BEV_t:<image> ego=center up=forward
collision_from={direction} in={time}s
N=20 dt=0.1
steer_deg in [-60,60], accel in [-9,9]
orig:<control sequence>
Before answering, reason step-by-step internally:
- interpret BEV motion and collision direction
- identify the primary avoidance strategy (brake,
  steer, or both)
- compute smooth per-step corrections respecting
  limits
Do NOT output reasoning.
Predict control adjustment CLASS to avoid collision.
Classes are discrete types of control changes (0-186).
Each class represents a combination of delta steer and
delta accel.
Predict control adjustment CLASS.
OUTPUT: class1,class2,...,class20
```

Training Objective

Given $\mathcal{P}$, the VLA autoregressively predicts a sequence of residual control classes. During SFT, a causal cross-entropy loss is applied over both the $\mathcal{R}$ and the final class outputs in Eq. (19), where y_ℓ denotes the ℓ-th target token in the output sequence, which includes both CoT tokens in $\mathcal{R}$ and the discrete residual control class tokens; $y_{<\ell}$ represents all preceding tokens in the autoregressive decoding process.

$$\mathcal{L} = -\sum_{\ell=1}^{L} \log p\left(y_\ell \mid y_{<\ell}, \mathcal{P}\right), \tag{19}$$

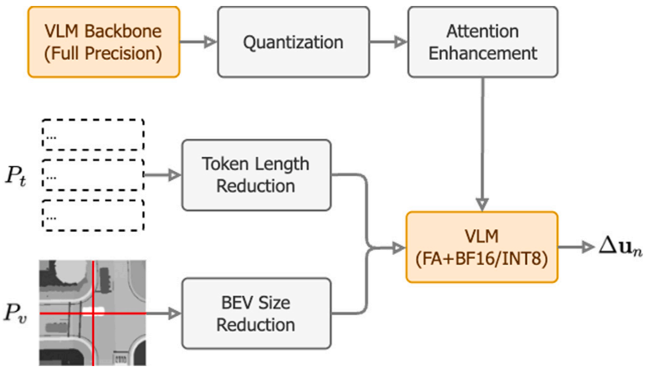

Fig. 3. Edge computing adaption process.

3.7. Residual trajectory fusion (RTF)

RTF is adopted not only for efficiency, but also because REACT operates as a hazard-triggered refinement module over an existing nominal planner. Under this setting, predicting residual control adjustments is more structurally suitable than generating a full control sequence from scratch, since it preserves a feasible nominal driving prior, reduces the difficulty of absolute sequence synthesis, and improves the stability and executability of the final control commands through constrained residual decoding and fusion.

From the reconstructed control sequence $\mathcal{U}$, residual control pairs $\Delta\mathbf{u}_n = (\Delta\delta_n, \Delta a_n)$ are collected and modeled using a GMM with K components, yielding a finite residual control vocabulary $\{c_1, \ldots, c_K\}$. During inference, the VLA predicts a sequence of residual control classes $\hat{\mathbf{c}}$ in Eq. (20).

$$\hat{\mathbf{c}} = \{\hat{c}_n\}_{n=1}^{N}, \quad \hat{c}_n \in \{1, \ldots, K\}. \tag{20}$$

Each $\hat{\mathbf{c}}$ corresponds to a Gaussian component in a pre-trained GMM over control residuals. Let the GMM be defined as Eq. (21), where $\boldsymbol{\mu}_k = (\mu_k^\delta, \mu_k^a)$ and $\boldsymbol{\Sigma}_k$ denote the mean and covariance of component k. Given a predicted class $\hat{c}_n$, the corresponding residual control is recovered by selecting the component mean, shown in Eq. (22), which yields a deterministic and stable residual $\Delta\mathbf{u}_n$.

$$p(\Delta\mathbf{u}) = \sum_{k=1}^{K} \pi_k\, \mathcal{N}\left(\Delta\mathbf{u} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\right), \tag{21}$$

$$\Delta\mathbf{u}_n = \mathbb{E}\left[\Delta\mathbf{u} \mid \hat{c}_n\right] = \boldsymbol{\mu}_{\hat{c}_n}, \tag{22}$$

Given a nominal control sequence $\mathbf{u}_{k+n}^{\text{nom}}$, RTF applies the decoded residuals with clipping as Eq. (23), producing a smooth, executable control sequence that preserves the original plan while adaptively mitigating collision risk. The optimized control commands are directly forwarded to the vehicle controller.

$$\hat{\mathbf{u}}_{k+n} = \text{Clip}\left(\mathbf{u}_{k+n}^{\text{nom}} + \Delta\mathbf{u}_n\right), \tag{23}$$

4. Edge computing adaptation

To enable practical deployment of REACT on resource-constrained edge platforms, this section introduces lightweight adaptation strategies that reduce computational and token overhead while preserving VLA reasoning fidelity and stable control prediction.

4.1. Adaptation overview

Fig. 3 illustrates the overall optimization pipeline for deploying REACT on edge devices. The adaptation process begins with the selection of an appropriate VLM backbone of the VLA model. SmolVLM2-500M is chosen due to its balance between compact model size and sufficient capability to accurately follow structured prompts. Larger models, like Qwen_2.5VL (3B and 7B) and LLaVA_1.5 (7B), exhibit prohibitively high latency, while smaller alternatives, such as SmolVLM2-256M, fail to consistently adhere to the required output format. Following model selection, this study performs quantization, converting the model weights from full precision (32-bit) to BF16 (16-bit) or INT8 (8-bit) precision to reduce memory usage and enhance inference efficiency. Next, the "Attention Enhancement" step incorporates scaled dot-product attention improvements, optimizing transformer self-attention calculations and further reducing computational overhead. To minimize inference latency, additional reductions in input complexity are implemented. In the "Token Length Reduction" step, the textual prompts are further reduced in token length to reduce the computation cost. Concurrently, the "BEV Size Reduction" step compresses the BEV feature map resolution, substantially decreasing the number of visual tokens required for inference.

4.2. Model quantization

Model quantization involves converting the model weights from full 32-bit floating-point precision to lower bit-depth precision, such as INT8 or FP16. This method reduces memory usage and computational demands by lowering the numerical precision required for storage and computation. Consequently, quantization enables faster inference and decreased power consumption, making real-time deployment feasible on edge devices without critically compromising model accuracy.

4.3. Computational complexity reduction

The attention enhancement strategy utilizes scaled dot-product attention (SDPA) to optimize the transformer's self-attention mechanism. By simplifying and streamlining matrix multiplication operations inherent in attention calculations, SDPA substantially decreases computational complexity. This approach improves the efficiency of inference operations, thereby enabling quicker and more responsive real-time predictions within computationally restricted environments.

4.4. Input size reduction

REACT reduces textual input size by adopting a compact instruction template optimized for VLA comprehension rather than human readability. Other key reductions include predicting residual controls instead of absolute sequences, encoding control plans as discrete numeric classes, rounding continuous values, and removing redundant or non-essential text while preserving task-critical semantics. These strategies shorten token length and improve output stability without sacrificing contextual understanding.

For visual inputs, BEV feature maps are compressed via channel-wise max pooling. This operation preserves salient road-user features while suppressing low-activation background pixels, reducing the visual token count, and mitigating noisy disturbances that can degrade prediction quality.

In addition, lightweight structured pruning is applied to the VLA backbone by removing low-importance attention heads and neurons, further reducing computational overhead while maintaining control prediction accuracy.

5. Experiment

5.1. Experimental setup

This study utilizes the *DeepAccident* dataset (Wang et al., 2024b), which comprises 691 accident scenarios simulated in CARLA (Dosovitskiy et al., 2017), grounded in real-world crash reports. The dataset provides comprehensive coverage of safety-critical driving contexts by integrating diverse weather and lighting conditions, varied traffic densities, and both crash and non-crash outcomes. It includes six high-risk intersection scenarios spanning signalized and unsignalized environments, and is evenly distributed across diverse geographical areas such as small towns, downtown apartment and commercial districts, rural roads, and highways. Each scenario offers multi-view camera and LiDAR data from vehicles and an RSU, supporting motion prediction and collision reasoning tasks, and enabling robust training of V2X-based collision avoidance models.

This paper constructs each data sample using two BEV observations: one from 1 s in the past and one at the current frame. Conditioned on the aligned hazard context, specifically the predicted accident's relative direction and time-to-event, the model predicts a future control sequence over a 2 s horizon. This formulation provides sufficient temporal context for short-horizon motion understanding while enabling direct, control-level prediction without relying on explicit future trajectory supervision. After preprocessing, the resulting training set consists of 10,125 valid samples for model fine-tuning. The validation set is preprocessed in the same way as the training set and contains 8827 samples for model evaluation.

To evaluate deployment performance on edge platforms, all fine-tuned models are tested on the NVIDIA Jetson AGX Orin (AGX Orin). All deployment experiments and all reported latency results in this work are based on this same hardware platform.

Hyperparameter tuning identifies a learning rate of 1×10^{-4} as optimal for training stability and convergence. Fine-tuning is conducted with a batch size of 6, a gradient accumulation step of 4, 25 warmup steps, and 20 epochs, resulting in a total of 33,751 steps across the complete training set.

This study evaluates various edge adaptation strategies, as detailed in Section 5.3. Based on the experimental outcomes, the optimal configuration utilizes 8-bit quantization, a text token length of 1500, and a BEV resolution of 90 × 90. Subsequent experiments adopt this optimized configuration to ensure optimal performance.

Beyond latency, the selected SmolVLM2-500M backbone is also favorable from a memory-efficiency perspective for embedded deployment. As a 0.5B-parameter model operated in the final 8-bit configuration, its parameter storage is substantially smaller than 3B- and 7B-class alternatives, leaving greater practical headroom for runtime buffers and surrounding driving software on Orin-class edge platforms. Since REACT functions as a hazard-triggered trajectory refinement module rather than the entire autonomy stack, this model scale is compatible with contemporary embedded automotive compute budgets.

5.2. Evaluation metrics

Following the evaluation protocol in Wang et al. (2024b), this study assesses model performance using three complementary metrics: Mean Intersection-over-Union (mIoU), Video Panoptic Quality (VPQ) (Hu et al., 2021), and Collision Rate Reduction (CRR).

Mean Intersection-over-Union (mIoU)

The mIoU measures the region-level overlap between predicted and ground-truth occupied areas, without distinguishing individual instances or temporal identities. For each accident scenario s, IoU is computed as Eq. (24), where $\hat{A}_s$ and A_s denote the predicted and ground-truth occupancy regions aggregated over the prediction horizon. The overall mIoU is then defined as Eq. (25). Compared to VPQ, mIoU reflects spatial alignment quality but does not capture instance separation or temporal consistency.

$$\mathrm{IoU}_s = \frac{|\hat{A}_s \cap A_s|}{|\hat{A}_s \cup A_s|}, \tag{24}$$

$$\mathrm{mIOU} = \frac{1}{S} \sum_{s=1}^{S} \mathrm{IoU}_s. \tag{25}$$

Table 2
Module-wise inference-time breakdown for 10-step (1 s) and 20-step (2 s) horizons, measured on the NVIDIA Jetson AGX Orin. Percentages are relative to the total inference time for each horizon.

Module (group)	10-step (ms)	10-step %	20-step (ms)	20-step %
BEV preprocessing				
BEV load + max-pool + edge-highlight	20.8	3.5%	20.8	1.4%
Control preprocessing				
Control reconstruction	0.04	<0.1%	0.07	<0.1%
Residual control classification (GMM, K=183)	5.7	1.0%	12.1	0.8%
VLA inference				
Vision encoding (embedding + encoder)	243.6	42.4%	606.8	41.8%
Text–vision fusion & LLM inference	256.2	44.6%	656.1	45.2%
Output decoding	48.2	8.4%	155.7	10.7%
Total	**574.5**	**100%**	**1451.6**	**100%**

Video Panoptic Quality (VPQ)

Following FIERY (Hu et al., 2021), VPQ evaluates future motion prediction by jointly measuring instance-level segmentation accuracy, recognition, and temporal consistency over a prediction horizon. Unlike frame-wise IoU metrics, VPQ explicitly accounts for false positives, false negatives, and identity mismatches across time.

At each future timestep $t \in \{0, \dots, H\}$, predicted instances p_t are matched to ground-truth instances q_t using Hungarian matching under an IoU threshold of 0.5. Let TP_t, FP_t, and FN_t denote the sets of true positive, false positive, and false negative instances at timestep t, respectively. VPQ is defined as Eq. (26). This formulation penalizes missed detections, spurious predictions, and temporal inconsistency, making VPQ particularly suitable for evaluating safety-critical, temporally coherent motion prediction.

$$\mathrm{VPQ} = \sum_{t=0}^{H} \frac{\sum_{(p_t,q_t)\in TP_t} \mathrm{IoU}(p_t, q_t)}{|TP_t| + \frac{1}{2}|FP_t| + \frac{1}{2}|FN_t|}. \tag{26}$$

Collision Rate Reduction (CRR)

To evaluate the effectiveness of REACT in mitigating collision risk, a frame-wise collision rate is computed over accident-related frames $f \in \{1, \dots, N\}$. Unlike point-based metrics, this evaluation measures the minimum physical separation between the ego vehicle and surrounding objects using their bounding boxes, providing a more realistic assessment of collision events.

For each frame f, let $\mathcal{B}_{\mathrm{ego}}^{(f)}$ denote the ego vehicle's bounding box and $\mathcal{B}_j^{(f)}$ denote the bounding box of surrounding object $j \in \mathcal{O}_{\mathrm{sur}}^{(f)}$. The minimum bounding-box distance $d_{\min}^{(f)}$ is defined as Eq. (27), where $\mathrm{dist}(\cdot,\cdot)$ computes the minimum Euclidean distance between the perimeters of two axis-aligned bounding boxes.

$$d_{\min}^{(f)} = \min_{j\in\mathcal{O}_{\mathrm{sur}}^{(f)}} \mathrm{dist}\left(\mathcal{B}_{\mathrm{ego}}^{(f)}, \mathcal{B}_j^{(f)}\right), \tag{27}$$

A collision is recorded for f if the minimum distance is less than 1 meter in Eq. (28). The collision rate C is then defined as Eq. (29).

$$F_{\mathrm{col}}^{(f)} = \begin{cases} 1, & \text{if } d_{\min}^{(f)} < 1 \text{ m} \\ 0, & \text{otherwise} \end{cases} \tag{28}$$

$$C = \frac{1}{N}\sum_{f=1}^{N} F_{\mathrm{col}}^{(f)} \tag{29}$$

Finally, the CRR achieved by REACT's collision rate C_{REACT} compared to the baseline's rate C_{baseline} is computed as Eq. (30).

$$\mathrm{CRR} = \frac{C_{\mathrm{baseline}} - C_{\mathrm{REACT}}}{C_{\mathrm{baseline}}} \tag{30}$$

5.2.1. Inference time analysis

Table 2 summarizes how inference latency is distributed across modules and prediction horizons. BEV preprocessing introduces a small, nearly constant overhead, as it only involves loading BEV features, spatial max-pooling, and edge-based highlighting on current and past observations, independent of horizon length.

Control-related preprocessing incurs negligible cost. Nominal control reconstruction relies on lightweight finite-difference operations, while residual control classification scales modestly with the horizon due to the increased number of GMM-encoded class predictions.

The dominant computational cost arises from the VLA inference stage. Vision encoding accounts for approximately 42% of total latency, and text–vision fusion with LLM inference contributes about 45%. Both scale almost linearly with the prediction horizon due to autoregressive decoding. Output decoding adds a smaller, consistent overhead. Overall, this breakdown shows that REACT's residual control formulation and compact class-based outputs effectively constrain horizon-dependent latency growth, enabling predictable scaling and real-time deployment.

5.2.2. Motion prediction comparison

Table 3 compares REACT with representative V2X fusion methods on the DeepAccident dataset using mIOU and VPQ. REACT is evaluated under identical experimental settings following Wang et al. (2024b), including the same BEV perception backbone, spatial resolution, 1-second past observation window, and a unified 2-second prediction horizon. Results for V2XFormer fusion variants and AccidentGPT results are cited from their original paper.

Traditional fusion methods rely on fixed-stage feature aggregation or Transformer-based fusion, while AccidentGPT introduces a general-purpose LLM for motion decoding. In contrast, REACT adopts a VLA formulation that jointly reasons over BEV perception and structured accident context to generate control-aware predictions. This unified design achieves the highest mIOU and a substantial VPQ improvement, outperforming all baselines under matched evaluation protocols.

The larger gain in VPQ than in mIoU is expected because the two metrics emphasize different aspects of prediction quality. While mIoU mainly reflects coarse region-level spatial overlap, VPQ is more sensitive to instance separation, false positives/negatives, and temporal consistency across the prediction horizon. REACT's cooperative hazard reasoning and control-aware prediction primarily improve the temporal coherence and behavioral plausibility of predicted motion, which leads to a stronger advantage in VPQ than in mIoU. For real driving, this suggests that REACT's main benefit lies less in marginally enlarging occupancy overlap and more in producing smoother, earlier, and more safety-relevant motion evolution for downstream planning and collision avoidance.

5.2.3. Comparison of VLM backbones

Table 4 summarizes motion prediction, CRR, and average inference latency on the NVIDIA Jetson AGX Orin for four VLM backbones. Larger models, like LLaVA-1.5-7B and Qwen2.5-VL-7B, obtain the highest raw motion accuracy, reflecting their stronger multimodal representation capacity. However, these accuracy gains come with substantial latency.

Table 3
Comparison of Motion Prediction Accuracy with Prior Fusion Methods. ↑: higher is better; bold value means the highest value; TF: Transformer.

Method	Perception Backbones	Fusion Backbones	Prediction Backbones	mIOU ↑	VPQ ↑
V2XFormer+Average Fusion (Wang et al., 2024b)	TF	Simple average	TF	52.1	39.5
V2XFormer+DiscoNet (Mehr et al., 2019)	TF	Autoencoder	TF	54.2	42.0
V2XFormer+V2X-ViT (Xu et al., 2022b)	TF	TF	TF	55.1	43.2
V2XFormer+CoBEVT (Xu et al., 2022a)	TF	TF	TF	56.2	44.0
AccidentGPT (Wang et al., 2023b)	TF	TF+VLM	LLM	57.3	45.2
REACT (ours)	TF	VLM	VLM	**57.9**	**50.6**

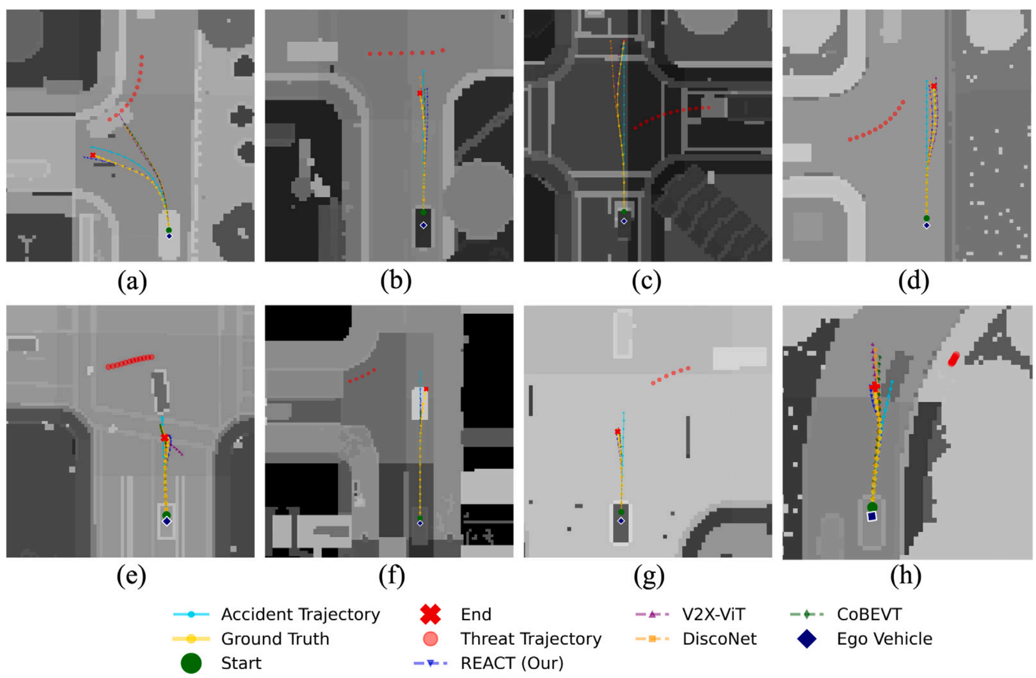

Fig. 4. Qualitative trajectory comparison in accident-prone scenarios. (a–d) Ego vehicle directly involved in different accident types, including intersection conflicts, curved-road collisions, and mid-block crashes. (e–h) Ego vehicle encountering secondary risks as a following vehicle. Compared with DiscoNet, V2X-ViT, and CoBEVT, REACT consistently exhibits earlier anticipation, smoother avoidance maneuvers, and safer stopping or steering behaviors.

Table 4
Comparison of different VLM backbones under identical training and evaluation settings. Motion metrics are reported at a 10-step (1 s) horizon. Higher CRR and lower inference time indicate better performance. Average inference time is measured on the NVIDIA Jetson AGX Orin.

Model	Motion (%) ↑		CRR (%) ↑	Avg Inference Time (s) ↑
	mIOU	VPQ	(GT=100)	AGX Orin
Qwen2.5-VL-3B	78.5	78.3	85.8	1.866
LLaVA-1.5-7B	**82.4**	79.8	86.3	2.938
Qwen2.5-VL-7B	80.1	**80.2**	**86.7**	2.090
SmolVLM2-500M	79.6	76.0	84.9	**0.570**

LLaVA averages 2.94 s and Qwen2.5-VL-7B about 2.09 s per inference, which is challenging for real-time edge deployment.

Regarding safety, the top absolute CRR in the table is achieved by Qwen2.5-VL-7B, followed closely by LLaVA-1.5-7B and Qwen2.5-VL-3B. SmolVLM2-500M attains a CRR of 84.9%, which is slightly lower than the largest models but remains high in absolute terms. Specifically, SmolVLM2-500M delivers this competitive CRR while offering an order-of-magnitude lower inference time on the AGX Orin, enabling a practical real-time response that the larger models cannot provide at comparable latency.

The results highlight a clear accuracy–latency–safety trade-off. Large VLMs provide marginally better motion metrics and the highest CRR, but at latencies that hinder edge deployment. The SmolVLM2-500M backbone represents a pragmatic middle ground. It preserves strong motion performance, achieves a high CRR, and meets real-time constraints on the target hardware. For REACT, whose objective is to enable timely, control-aware collision mitigation on resource-limited edge devices, this paper therefore favors SmolVLM2-500M as the backbone due to its balanced combination of safety, accuracy, and real-time efficiency.

5.2.4. Qualitative results

Fig. 4 presents qualitative comparisons across eight representative accident-prone scenarios. Subplots (a)–(d) depict cases where the ego vehicle is directly involved in different accident types, while (e)–(h)

Table 5
Collision Rate Reduction (CRR, %) of REACT under different environmental and spatial conditions.

Condition	Category	CRR (%)
Time of Day	Noon	85.01
	Sunset	84.88
	Night	84.90
Weather	Clear	85.12
	Cloudy	84.76
	Rainy	84.81
	Wet	84.83
Location	Highway	84.70
	Downtown	84.65
	Residential	85.10
	Rural	85.00
Overall Average	–	**84.93**

Table 6
Comparison of model variants (10-step horizon = future 1 s trajectory). All inference times are measured on the Jetson AGX Orin.

Model	CRR %	mIoU %	VPQ %	Inference Time@10-step (s)	Inference Time@20-step (s)
SFT	85.96	57.83	50.61	1.43	2.68
RTF + SFT	84.51	56.72	49.24	0.60	1.58
RTF + CoT-SFT	84.93	57.85	50.59	0.57	1.45

show scenarios where the ego vehicle encounters secondary risks as a following vehicle.

In direct-accident cases, baseline methods often react late or follow near-straight trajectories that intersect with accident regions. In (a), at a T-intersection conflict, DiscoNet and V2X-ViT initiate avoidance only after entering the conflict zone, whereas REACT steers earlier to maintain clearance. In curved-road scenarios (b) and (d), baselines exhibit delayed lateral deviation, while REACT performs smoother, earlier adjustments. In the mid-block collision case (c), REACT halts before reaching the crashed vehicle, unlike other methods that approach dangerously close.

Secondary-risk scenarios further highlight REACT's advantage. In (e) and (f), baselines continue forward until the accident becomes visible, while REACT decelerates earlier based on inferred downstream risk. In (g) and (h), REACT avoids cascading rear-end collisions by proactively stopping or adjusting its heading, demonstrating more conservative and anticipatory behavior.

These examples show that REACT generates safer, smoother, and more foresighted trajectories by reasoning over accident context and surrounding dynamics, outperforming existing V2X fusion baselines in both direct and secondary accident scenarios.

5.2.5. REACT performance under various conditions

Table 5 reports the CRR achieved by REACT across different times of day, weather conditions, and location types. Across all evaluated settings, CRR remains tightly clustered between 84.6% and 85.1%, with an overall average of 84.93%. This narrow variance indicates that REACT exhibits strong robustness and generalization, maintaining consistent collision mitigation performance despite changes in visibility, road surface conditions, and traffic structure.

Slightly higher CRR is observed in residential and rural areas, where lower speeds provide greater maneuvering margins, while highway and downtown scenarios show marginally lower, but still strong performance due to higher speeds and denser traffic. Similarly, nighttime and adverse weather conditions introduce only minor degradation. Overall, these results demonstrate that REACT's control-aware VLA reasoning generalizes well across diverse operational conditions, supporting reliable real-world deployment.

5.3. Ablation studies

This section presents ablation experiments to evaluate the impact of three key design choices in REACT: (1) the inclusion of the RTF module and the choice of CoT-SFT, (2) the influence of edge adaptation strategies, and (3) the influence of model inputs.

5.3.1. Effect of RTF and CoT-SFT

To empirically examine the contribution of residual control formulation and language-based CoT supervision, this paper compares three model variants: (i) *SFT*, which directly predicts the full control sequence without RTF; (ii) *RTF + SFT*, which predicts discrete residual control classes without CoT; and (iii) *RTF + CoT-SFT*, which further incorporates CoT supervision during fine-tuning.

As shown in Fig. 7 and Table 6, the SFT model achieves the highest raw prediction accuracy, with mIoU/VPQ of 57.83/50.61 at the 10-step horizon. However, this accuracy comes at a high computational cost, with inference times of 1.43 s and 2.68 s for 10- and 20-step predictions, which are impractical for real-time deployment.

Introducing RTF reduces inference time by more than 58% at 10 steps and 41% at 20 steps, at the cost of a moderate accuracy drop. With CoT supervision, RTF + CoT-SFT recovers most of this performance loss, achieving 57.85/50.59 mIoU/VPQ, comparable to SFT, while further reducing inference time to 0.57 s and 1.45 s. This corresponds to over 60% latency reduction relative to SFT with less than 0.1% absolute accuracy difference.

Fig. 5 reports steering and acceleration prediction errors for three variants. The SFT model achieves the lowest median MAE and RMSE, reflecting its direct supervision on full control sequences. Introducing RTF increases both steering and acceleration errors, as residual control classification constrains outputs to a discrete action space. However, incorporating CoT supervision reduces this error inflation. Compared with RTF+SFT, RTF+CoT-SFT consistently lowers median errors and tightens interquartile ranges, particularly for steering RMSE, indicating more stable and coherent control adjustments. These results show that CoT guidance improves the reliability of residual control prediction, mitigating discretization-induced error while preserving the efficiency benefits of RTF.

These results demonstrate that RTF module and CoT supervision provide a favorable accuracy–latency balance, enabling reasoning-guided control prediction suitable for real-time autonomous driving applications.

5.3.2. Effect of model size and prune rate

Fig. 6 illustrates the joint impact of VLA model size and pruning rate on inference time and VPQ. The 256M model achieves low inference latency of approximately 0.5 s, but its VPQ remains around 23 percent, indicating insufficient representational capacity to reliably capture structured prompts and multimodal driving context. In contrast, the 500M model achieves substantially higher VPQ, reaching approximately 48 percent without pruning, while maintaining moderate inference latency of about 0.55 s.

As the pruning rate of the 500M model increases from 0 percent to 30 percent, inference time remains nearly unchanged, varying only slightly around 0.55 to 0.6 s, while VPQ gradually decreases from approximately 48 percent to 46 percent. When the pruning rate reaches 40 percent, VPQ further drops to roughly 45 percent, and inference time increases sharply to over 2 s. This behavior suggests that excessive pruning disrupts efficient hardware execution, leading to degraded performance despite reduced parameter count. Overall, the results indicate that moderate pruning preserves both accuracy and efficiency, whereas aggressive pruning is counterproductive, motivating the adoption of a lightly pruned 500M VLA model in REACT.

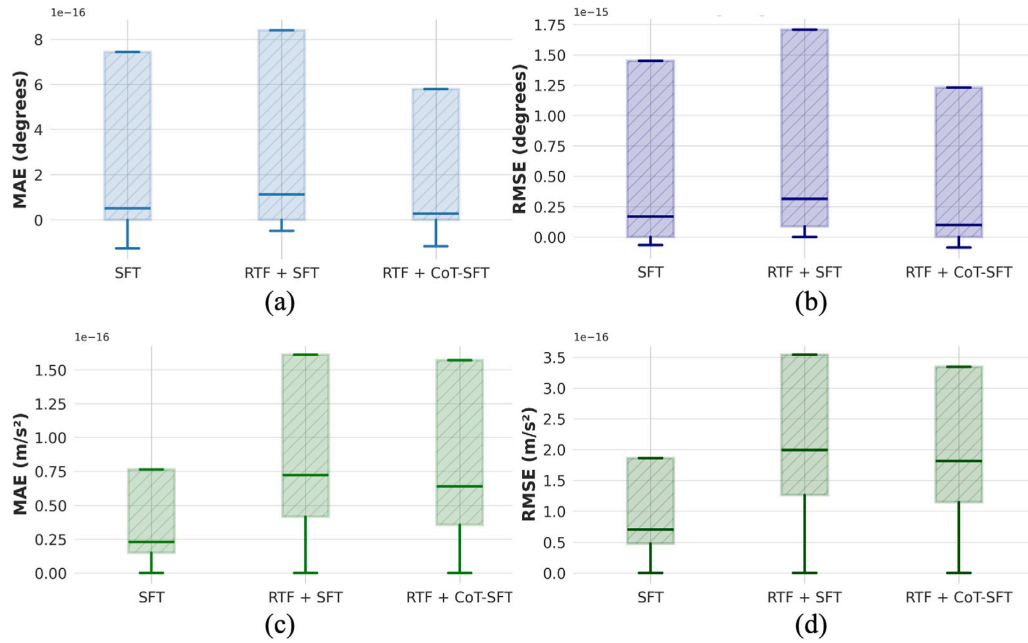

Fig. 5. Distribution of control prediction errors across model variants. (a) Steering angle MAE, (b) steering angle RMSE, (c) acceleration MAE, and (d) acceleration RMSE.

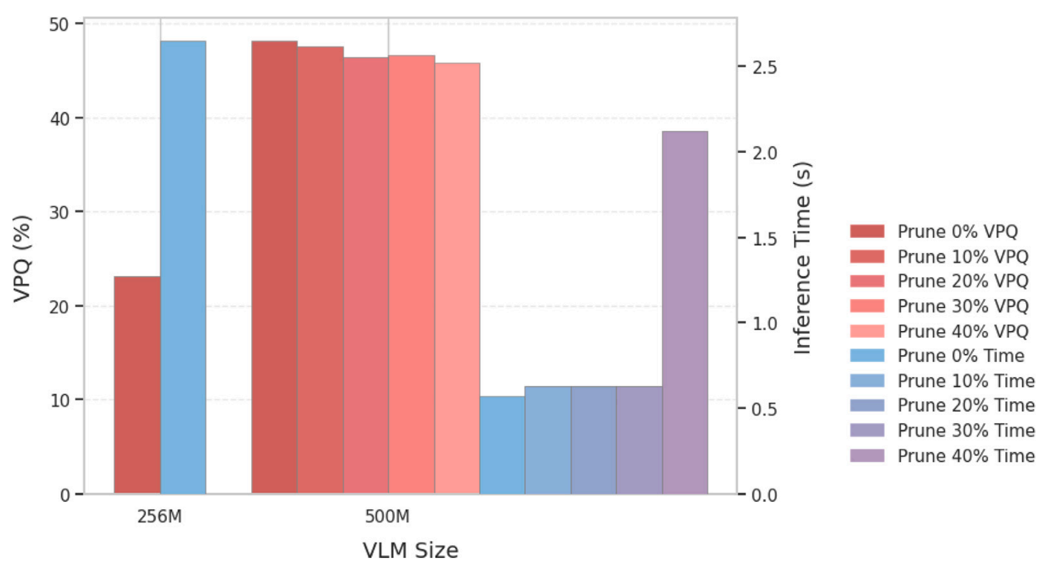

Fig. 6. Impact of BEV size and pruning on VPQ and inference time.

Table 7
Analytic ablation of edge adaptation strategies at a 10-step (1 s) prediction horizon. All values are reported relative to the reference configuration (8-bit quantization, ∼1500 tokens, 720 × 720 BEV resolution, SDPA).

Change from Reference	Quant.	Tokens	BEV Size	ΔTime (s)	ΔVPQ (%)
Reference configuration	8-bit	∼1500	90 × 90	0.00	0.00
Increase token length	8-bit	∼2200	90 × 90	+1.16	+1.25
Increase BEV resolution	8-bit	∼1500	720 × 720	+0.02	+0.56
Without SDPA	8-bit	∼1500	90 × 90	+0.01	+0.10
Switch to 16-bit quantization	16-bit	∼1500	90 × 90	+0.04	+0.57
High-cost configuration	16-bit	∼2200	720 × 720	+1.18	+1.81

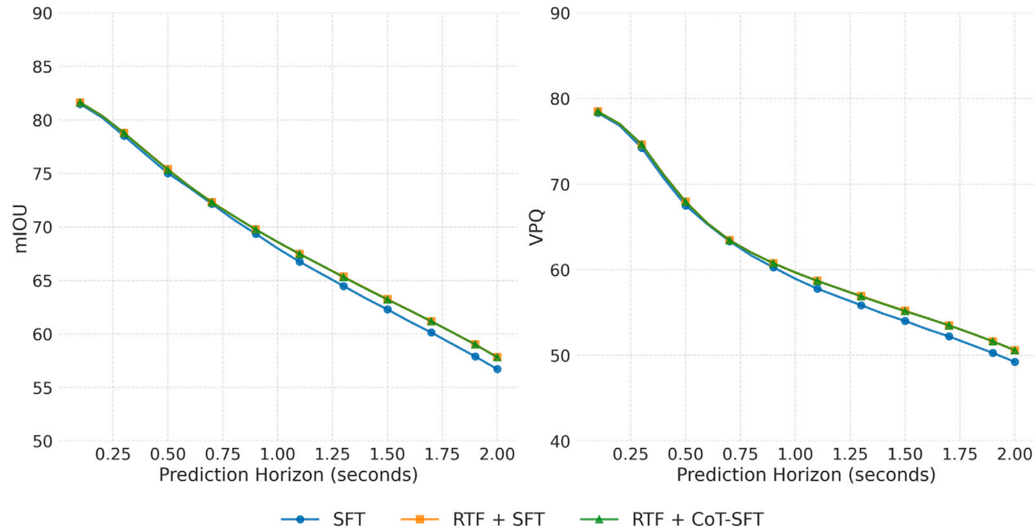

Fig. 7. Visualization of Vehicle Motion Prediction Metrics over 2 s horizon.

Table 8
Ablation study on input modalities. ↑: higher is better; ↓: lower is better.

Input removed	VPQ ↑	mIoU ↑	CRR (%) ↑
Visual frames	45.8 (−4.8)	54.6 (−3.3)	61.2 (−23.7)
RSU Hazard	46.4 (−4.2)	53.9 (−4.0)	67.1 (−17.8)
Control Sequence	32.2 (−18.4)	38.2 (−19.6)	51.7 (−33.2)

5.3.3. *Effect of edge adaption strategies*

Table 7 presents an analytic ablation of edge adaptation strategies relative to the reference configuration using 8-bit quantization, approximately 1500 tokens, a 90 × 90 BEV resolution, and SDPA, with inference time 0.57 s and VPQ 50.59. Overall, input token length dominates inference latency, while BEV resolution and quantization primarily affect accuracy–efficiency trade-offs.

The behavior is influenced by the specific software environment, where JetPack 6.x with CUDA 12.2 and TensorRT 8.6 provides more mature and optimized FP16 kernels for transformer attention and multimodal operations than for INT8 in some execution paths. As such, INT8 quantization does not always guarantee superior speed on this platform due to the current kernel optimization landscape in TensorRT and CUDA.

Impact of Token Length Increase.

Token increase corresponds to replacing GMM-encoded residual control classes with raw continuous control sequences and higher-precision numerical fields. This substantially increases transformer token count, leading to over 1 s additional latency with only modest VPQ improvement.

Impact of BEV Resolution.

Increasing BEV resolution from 90 × 90 to 720 × 720 yields a small VPQ gain at minimal latency cost, indicating that pooled BEV representations already preserve most safety-critical spatial information.

Impact of Attention Enhancement.

Removing SDPA consistently increases inference time with negligible VPQ change, confirming SDPA as an effective and nearly cost-free optimization for multimodal transformer attention.

Impact of Quantization Precision.

Switching from 8-bit to 16-bit quantization slightly increases latency but improves VPQ, reflecting better kernel optimization for FP16 multimodal transformer execution on the target edge platform.

These results motivate the final design choice of compact tokenization, moderate BEV resolution, SDPA, and lightweight quantization to balance real-time efficiency and predictive quality.

5.3.4. *Effect of inputs and modules*

Table 8 quantifies the contribution of key input modalities to REACT's motion prediction accuracy and collision mitigation performance by selectively removing individual inputs from the full model. The results show that while different inputs affect motion metrics to varying degrees, their impact on CRR differs substantially, highlighting the importance of multimodal, control-aware reasoning.

Visual Frames.

Removing the camera-derived BEV inputs causes a pronounced degradation across all metrics. VPQ and mIoU drop by 4.8 and 3.3 points, respectively, while CRR decreases sharply from 84.93% to 61.2%. This degradation reflects the loss of not only instantaneous spatial context, such as lane geometry and relative positioning of surrounding vehicles, but also short-horizon temporal cues encoded by the historical BEV that are critical for inferring motion trends and intent. Without this spatiotemporal visual context, REACT struggles to localize hazards accurately, anticipate dynamic interactions, and maintain safe clearances. These results confirm that paired current–historical BEV representations are fundamental to both accurate motion prediction and safe, anticipatory control generation.

RSU Hazard Input.

Ablating the RSU-provided hazard information also results in a significant performance drop, particularly in safety. While motion metrics decline moderately, CRR falls to 67.1%, reflecting a 17.8-point reduction. This demonstrates the critical role of infrastructure assistance in enabling proactive behavior. Without explicit downstream hazard alerts, REACT must rely solely on onboard perception, which limits its ability to anticipate occluded or distant accidents and reduces its effectiveness in early collision avoidance.

Control Sequence Input.

Removing the nominal control sequence leads to the most severe degradation among all ablations. Motion accuracy collapses, and CRR drops dramatically to 51.7%. This highlights the central role of the control sequence in grounding the VLA's reasoning process. Without a nominal plan, residual control prediction becomes ill-posed, preventing REACT from generating stable and meaningful corrective actions. This result validates the design choice of residual trajectory fusion, where control-aware inputs are essential for both accurate prediction and safe execution.

6. Conclusion and discussion

This paper proposes REACT, a real-time V2X-enabled trajectory planning framework based on a lightweight VLA model. By integrating cooperative hazard alerts, spatiotemporal BEV perception, and control-aware reasoning, REACT directly generates executable control adjustments and enables proactive collision avoidance on resource-constrained edge platforms.

Extensive experiments on the DeepAccident benchmark demonstrate the effectiveness of REACT across accuracy, safety, and efficiency dimensions. REACT achieves competitive motion prediction performance, 57.85% for mIoU and 50.59% for VPQ, while reducing collision rates by 84.93% relative to the baseline. Compared with existing V2X fusion methods and larger VLM backbones, REACT maintains strong safety performance with lower inference latency, achieving real-time execution at 0.57 s on the NVIDIA Jetson AGX Orin. These results validate that control-aware VLA reasoning, rather than purely improving motion accuracy, is critical for practical and safety-oriented autonomous driving deployment. Key findings across specific capabilities are summarized below:

Contextual Reasoning

REACT effectively leverages cooperative V2X signals and spatiotemporal BEV representations within a finetuned VLA model to reason about occluded hazards and multi-agent interactions. Both quantitative results and qualitative examples show earlier anticipation of downstream risks and more stable, proactive control adjustments than baseline methods.

Multimodal Fusion

By jointly integrating paired BEV visual inputs with structured symbolic information such as hazard direction, time-to-event, and nominal control plans, REACT achieves strong VPQ and mIoU performance. Input ablations confirm that removing either visual or symbolic modalities substantially degrades both motion accuracy and safety, highlighting their complementary roles.

Edge Deployment

REACT is explicitly designed for real-time edge execution. Through lightweight VLM selection, residual control prediction, quantization, attention optimization, and compact prompt design, REACT achieves a 0.57 s inference latency on the NVIDIA Jetson AGX Orin while maintaining strong motion accuracy and safety performance, demonstrating practical deployability on a resource-constrained edge platform. This deployability advantage arises not only from reduced latency, but also from the compact memory footprint of the selected 500M, 8-bit backbone relative to larger multimodal models, which improves suitability for Orin-class edge platforms used in real-time driving applications.

Generalization

Across different times of day, weather conditions, and location types, REACT consistently maintains a high collision rate reduction with minimal variance. This stability indicates that its control-aware VLA reasoning generalizes well beyond specific scenarios, supporting reliable operation under diverse and previously unseen driving conditions.

Ethical and Safety Considerations

Because REACT directly refines low-level steering and acceleration commands, its real-world deployment requires careful safety considerations. The system must remain reliable under V2X communication delays, packet loss, RSU disconnection, and noisy or incorrect hazard messages, which should be mitigated through validation and fallback control policies. Although residual control prediction and clipping improve stability, AI-driven decisions should be traceable to support auditing and human oversight in safety-critical scenarios.

Despite REACT's strong performance, several limitations warrant further investigation. First, the current evaluation is conducted exclusively on the DeepAccident benchmark within a CARLA-based simulation environment using a fixed planning stack, and therefore does not yet demonstrate cross-simulator, cross-dataset, or real-world generalization. In addition, the use of preprocessed BEV representations does not fully capture perception noise, sensing uncertainties, communication imperfections, or closed-loop feedback effects that arise in real deployments. Second, although real-time inference is achieved on edge hardware, additional latency reductions may be possible through more advanced optimization of transformer attention mechanisms and decoding strategies. Finally, further improvements in collision rate reduction may depend on enhancing motion prediction and control fidelity, which in turn may require higher-quality and more precise control and hazard annotations in existing datasets.

Future research will focus on advancing REACT toward robust, transparent, and real-world deployable autonomous driving systems. Beyond the current single-simulator evaluation, an important next step is to assess REACT across different simulators, datasets, and planning stacks to better understand its deployment boundaries. The research group also plans to conduct systematic comparisons with prior residual-based trajectory refinement and planning approaches by adapting them to accident-centric V2X benchmarks, enabling clearer contextualization of RTF's contribution. Integrating REACT into a closed-loop system with online perception, planning, and actuation will allow evaluation under realistic sensing noise, V2X communication latency, packet loss, and localization uncertainty, while additional robustness analyses with controlled noise injection and delayed hazard cues will further quantify safety margins. At the same time, as VLA-based systems directly influence control decisions, future work will emphasize robustness under degraded communication, improved decision transparency, and safety assurance mechanisms to address concerns related to accountability and public trust. Finally, continued optimization of transformer attention, decoding, and model compression will be explored to further reduce latency and support scalable, safety-critical edge deployment.

7. Glossary

V2X (Vehicle-to-Everything): A cooperative communication paradigm that enables vehicles to exchange information with other vehicles, roadside units (RSUs), infrastructure, and networks to extend perception beyond onboard sensing and support proactive safety reasoning.

Vision–Language–Action Model (VLA): A multimodal model that jointly processes visual inputs, structured symbolic information, and language-based prompts to directly generate executable low-level actions, such as steering and acceleration commands.

Vision–Language Model (VLM): A multimodal neural architecture that fuses visual representations and textual inputs to enable semantic understanding and contextual reasoning across modalities.

Edge Computing: A decentralized computing paradigm in which inference and decision-making are performed close to the data source (e.g., onboard vehicles or RSUs) to reduce latency and support real-time, safety-critical applications.

Ego Vehicle: The autonomous vehicle under control and evaluation, serving as the reference frame for perception, reasoning, and control generation.

Residual Trajectory Fusion (RTF): A control reconstruction module that predicts discrete residual control adjustments and fuses them with a nominal control sequence to generate stable, executable actions with reduced inference latency.

Chain-of-Thought Supervised Fine-Tuning (CoT-SFT): A training strategy that supervises intermediate reasoning steps during fine-tuning to improve task understanding, output stability, and generalization, while suppressing reasoning traces at inference.

Bird's-Eye View (BEV): An ego-centric, top-down spatial representation of the driving environment constructed from sensor data, enabling unified spatial and temporal reasoning over road geometry and surrounding agents.

Video Panoptic Quality (VPQ): A spatiotemporal evaluation metric that jointly measures instance-level segmentation accuracy, recognition, and temporal consistency for future motion prediction.

mIoU (Mean Intersection-over-Union): A region-level metric that measures spatial overlap between predicted and ground-truth occupancy areas, without accounting for instance identity or temporal consistency.

Collision Rate Reduction (CRR): A safety-oriented metric that quantifies the relative reduction in collision occurrence achieved by a method compared to a baseline, based on minimum bounding-box distances over time.

Prompt Engineering: The design of structured multimodal prompts that encode task objectives, environmental context, and control constraints to guide VLA inference efficiently and reliably.

Quantization: A model optimization technique that reduces numerical precision (e.g., FP16 or INT8) to lower memory footprint and computational cost for edge deployment.

Scaled Dot-Product Attention (SDPA): An optimized transformer attention mechanism that improves computational efficiency and inference speed, particularly on modern GPU-based edge platforms.

C-V2X (Cellular Vehicle-to-Everything): A cellular communication standard that supports low-latency, high-reliability V2X message exchange between vehicles and infrastructure.

Jetson AGX Orin: An NVIDIA edge AI computing platform used in this work to evaluate real-time inference performance of REACT under resource-constrained deployment conditions.

CRediT authorship contribution statement

Fengze Yang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Bo Yu:** Writing – review & editing, Visualization, Validation, Software. **Yang Zhou:** Writing – review & editing, Resources. **Xuewen Luo:** Visualization, Software. **Zhengzhong Tu:** Writing – review & editing, Resources. **Chenxi Liu:** Writing – review & editing, Validation, Supervision, Resources, Investigation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used OpenAI's GPT-4o language model solely in order to polish grammar and improve the clarity of writing during manuscript preparation. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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Declaration of competing interest

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Data availability

The data that support the findings of this study are available from the DeepAccident dataset (Wang et al., 2024b), which can be accessed at https://deepaccident.github.io. Processed data and code supporting the conclusions of this article are available from the corresponding author upon reasonable request.

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